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REVIEW ARTICLE

State-of-the-art of battery state-of-charge determination

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Abstract

From the early days of its discovery, humanity has depended on electricity, a phenomenon without which our technological advancements would not have been possible. With the increased need for mobility, people moved to portable power storage—first for wheeled applications, then for portable and finally nowadays wearable use. Several types of rechargeable battery systems, including those of lead–acid, nickel–cadmium, nickel–metal hydride, lithium ion and lithium-ion polymer exist in the market. The most important of them will be discussed in this review. Almost as long as rechargeable batteries have existed, systems able to give an indication about the state-of-charge (SoC) of a battery have been around. Several methods, including those of *direct measurements, book-keeping* and *adaptive systems* (Bergveld *et al* 2002 *Battery Management Systems, Design by Modelling* (*Philips Research Book Series*) vol 1 (Boston: Kluwer)) are known in the art for determining the SoC of a cell or battery of cells. An accurate SoC determination method and an understandable and reliable SoC display to the user will improve the performance and reliability, and will ultimately lengthen the lifetime of the battery. However, many examples of poor accuracy and reliability can be found in practice (Bergveld *et al* 2002, cited above). This review presents an overview on battery technology and the state-of-the-art of SoC methods. The goal of all the presented SoC indication methods is to design an SoC indication system capable of providing an accurate SoC indication under all realistic user conditions, including those of spread—in both battery and user behaviour, a large temperature and current range and ageing of the battery.

Keywords: battery, portable energy, adaptive systems, capacity monitoring, run-time, state-of-charge, state-of-health

(Some figures in this article are in colour only in the electronic version)

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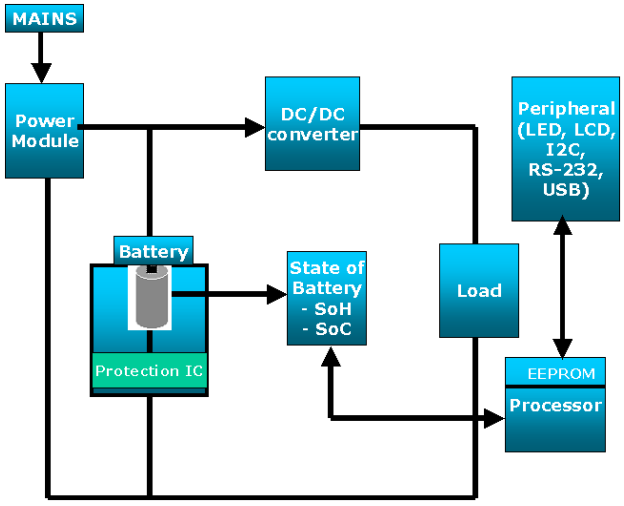


Figure 1. General architecture of a battery-management system [1].

1. Introduction

Portable electronic devices have become ubiquitous in modern society. The recent rapid expansion in the use of portable computers, personal data assistants, cellular phones, camcorders and power tools creates a strong demand for fast deployment of battery technologies at an unprecedented rate. The design of such a portable device requires many battery-management features, including charge control, battery capacity monitoring, remaining run-time information, compensation of battery capacity and remaining run-time due to environmental conditions, charge-cycle counting, etc.

An accurate SoC determination method and an understandable and reliable SoC display to the user will improve the performance and reliability, and will ultimately lengthen the lifetime of the battery. Accurate SoC information allows the battery to be used within the design limits, so the pack does not need to be over-engineered. This allows a smaller, lighter battery, which costs less. However, many examples of poor accuracy and reliability can be found in practice. This can be pretty annoying, especially when a portable device suddenly stops functioning whereas sufficient battery capacity is indicated. A poor reliability of the SoC indication system may induce the use of only part of the available battery capacity. For example, the user may be inclined to recharge the battery every day, even when enough battery capacity is indicated on the portable device. This will lead to more frequent recharging than strictly necessary, which in turn leads to an earlier wear-out of the battery. The effect of inaccuracy of SoC indication can be even worse when the SoC value is also used to control charging. The battery is either not fully charged or it is overcharged. In the former case, the battery will be recharged more often than needed, which will lead to an earlier wear-out. In the latter case, frequent overcharging will lead to a lower cycle life [1]. These examples motivate the request for an accurate and reliable SoC indicator system in portable applications. This review deals with this important part of the battery-management system (BMS).

A general block diagram of a BMS is shown in figure 1. The basic task of the power module (PM) is to charge the battery by converting electrical energy from the mains into electrical energy suitable to charge the battery (see figure 1). The PM can either be a separate device, such as a travel charger, or be integrated within the portable device, as in for example shavers [1]. A protection IC connected in series with the battery is needed for Li-ion batteries. The reason for this is that battery suppliers are particularly concerned with safety issues due to liability risks. The battery voltage, current and temperature have to be monitored and the protection IC has to be controlled to ensure that the battery is never operated in an unsafe region. The battery manufacturer determines the region within which it is considered safe to use the Li-ion battery. Outside the safe region, destructive processes may start to take place [1].

The dc/dc converter is used to efficiently condition the unregulated battery voltage (3–4.2 V in Li-ion chemistry) for compatibility with stringent load requirements (see figure 1). The basic task of the load is to convert the electrical energy supplied by a battery into an energy form that will fulfil the load's function, such as mechanical energy, light, sound, heat, EM radiation, etc.

The battery status can be indicated in a single or a string of light-emitting diode(s) (LED) or on a liquid-crystal display (LCD) that indicates the SoC and the battery condition (e.g., the state-of-health (SoH)) [1]. The processor is used to run the battery-management software, including the SoC algorithm (see figure 1). The communication between the BMS and other devices is another important task of the BMS. Depending on the application different systems for data exchange can be used, such as an inter-integrated-circuit bus interface (I^2C) or some other form of serial interface (see figure 1).

The battery state is used as an input parameter for the portable device electrical management and additionally it is an important parameter for the user. The battery state can be used to estimate the expected lifetime of the battery. It can simply be described by two parameters: SoC and SoH. Both parameters depend on each other and influence the battery performance. More explanations about this dependence will be given in section 5.

This review is organized as follows. Section 2 briefly describes the general operational mechanism of batteries and the characteristics of the best-known batteries with their applications. The history of SoC indication is presented in section 3. A general state-of-charge system and possible state-of-charge indication methods are discussed in sections 4 and 5, respectively. Section 6 focuses on commercially available state-of-charge systems. Finally, section 7 presents the concluding remarks.

2. Battery technology and applications

From the early days of electricity on, humanity has become dependent on electricity, a product without which our technological advancements would not have been possible. With the increased need for mobility, people moved to portable power storage—first for wheeled applications, then for portable and finally wearable use. As awkward and unreliable as the early batteries may have been, our descendants may one day look at today's technology in a similar way to how we view our predecessors' clumsy experiments of 100 years ago [2]. Improvements over time in the use of electricity, and in

particular battery technology as well as SoC indication will be presented in this review. This section focuses on developments in battery technology and characteristics.

Batteries have been around for a long time. In 1800, Volta discovered that a continuous flow of electrical force was generated when using certain fluids as conductors to promote an electrochemical reaction between two metals or electrodes [2]. This led to the invention of the first voltaic cell, better known as the battery.

Later in 1859, the French physicist Gaston Planté invented the first rechargeable battery. This secondary battery was based on *lead–acid (LA) chemistry*, a system that is still used today. In 1899, Waldmar Jungner from Sweden invented the *nickel–cadmium (NiCd) battery*, which used nickel for the positive electrode and cadmium for the negative. Two years later, Edison produced an alternative design by replacing cadmium with iron. Due to high material costs compared to dry cells or LA storage batteries, the practical applications of the nickel–cadmium and nickel–iron batteries were limited. It was not until Schlecht and Ackermann invented the sintered pole plate in 1932 that large improvements were achieved. These advancements were reflected in higher load currents and improved longevity. The sealed nickel–cadmium battery, as known today, only became available when Neumann succeeded in completely sealing the cell in 1947 [2].

Soon after the discovery in the late 1960s that intermetallic compounds, such as $SmCo_5$ and $LaNi_5$, were able to absorb and also desorb large amounts of hydrogen [3], it was realized that electrodes made of these materials could serve as a new electrochemical storage medium [4, 5]. In the following years the hydride-forming electrode has proven to be a serious alternative to the cadmium electrode, which was widely employed in rechargeable nickel–cadmium batteries. Particularly, the higher energy storage capacity, good rate capability and the non-toxic properties of the chemical elements from which these hydride-forming materials were composed were of benefit compared to the cadmium electrode [6]. The *nickel–metal hydride (NiMH) battery* became commercially available in the 1990s [7].

The first non-rechargeable *lithium batteries* appeared in the early 1970s. Attempts to develop rechargeable lithium batteries followed in the 1980s but failed due to safety problems. Because of the inherent instability of lithium metal, especially during charging, research shifted to a non-metallic lithium battery using lithium ions. Although lower in energy density than lithium metal, lithium ion is safe, provided certain precautions are met when charging and discharging, implemented by means of a safety IC in series with the battery as discussed in the previous section. In 1991, the Sony Corporation commercialized the first *lithium-ion battery (Li-ion)*.

The general operational mechanism of a battery and characteristics of the most important rechargeable batteries available in the market today, e.g. *nickel–cadmium, nickel–metal hydride* and *lithium-ion batteries*, will be given in the remainder of this section.

2.1. General operational mechanism of batteries

In its simplest definition, a battery is a device capable of converting chemical energy into electrical energy and vice versa. The chemical energy is stored in the electro-active species of the two electrodes inside the battery. The conversions occur through electrochemical reduction–oxidation (redox) or charge-transfer reactions [1]. These reactions involve the exchange of electrons between electro-active species in the two electrodes through an electrical circuit external to the battery. The reactions take place at the electrode/electrolyte interfaces. When current flows through the battery, an oxidation reaction will take place at the anode and a reduction reaction at the cathode. The oxidation reaction yields electrons to the external circuit, while a reduction reaction takes up these electrons from the external circuit. The electrolyte serves as an intermediate between the electrodes. It offers a medium for the transfer of ions. Hence, current flow is supported by electrons inside the electrodes and by ions inside the electrolyte. Externally, the current flows through the charger or load [1]. The basic electrochemical unit of a battery is called a *cell*, but commonly the word *battery* is used for one cell or for two or more cells connected in series/parallel.

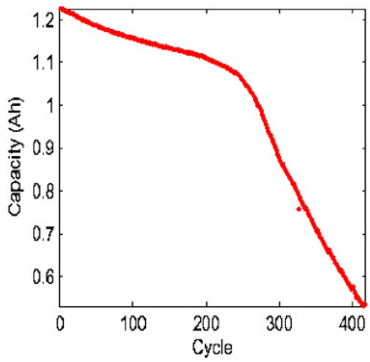


Figure 2. Capacity loss during cycling of a CGR18500 battery.

During the lifetime of a battery, its performance or 'health' tends to deteriorate gradually due to irreversible physical and chemical changes that take place with usage and with age until eventually the battery is no longer usable. The SoH is an indication of the point that has been reached in the life cycle of the battery and a measure of its condition relative to a fresh battery. Ageing of the battery is a complex process that involves many parameters of the battery (e.g., impedance, conductance, capacity) of which the most important is the capacity of the battery. To illustrate the phenomena let us look at figure 2 where the discharge capacity of a Li-ion battery is represented as a function of the cycle number. As we can see the degradation curve has a clearly visible change point, where the rate of degradation of the battery increases.

The exact place of the change point varies, depending on the type of battery and operational conditions. The ageing of Li-ion batteries is not an absolutely new topic in modern electrochemistry. Recently, a number of models describing the ageing effects in Li-ion batteries have been introduced. Models describing the dynamics of lithium consumption in Li-ion batteries are considered, e.g., in [8, 9]. Both Broussely and Spotnitz deal with the ageing of the batteries under 'on float' condition, where the battery is kept under constant voltage of fixed polarity. A striking feature of all plots in the mentioned articles is that they are smooth and do not have change points.

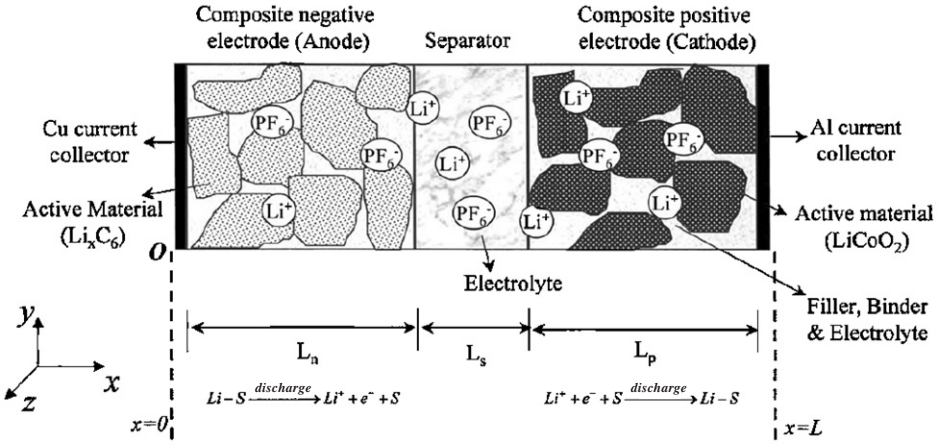


Figure 3. A schematic representation of a typical Li-ion cell [10].

A first conclusion is that it will be difficult to take into account every nuance of a battery's charge and discharge qualities and the ageing characteristics.

2.2. Battery types and characteristics

2.2.1. NiCd batteries. Batteries which use a positive nickel electrode of a $Ni(OH)_2/NiOOH$ compound and a negative cadmium electrode of Cd and $Cd(OH)_2$ are called NiCd batteries. The electrolyte is an aqueous KOH solution. Advantages of the NiCd batteries are their fast charge and discharge times: it is possible to charge a battery in 10 min and large currents can be supplied. NiCd batteries have a cell voltage of 1.2 V and can be used in many devices. Especially in tools demanding much power the NiCd battery is often found. Other applications include cordless and mobile phones, shavers, camcorders, portable audio products and laptop computers. Disadvantages of the NiCd batteries are their low-energy density and their so-called memory effect. The memory effect causes the battery to deliver only the capacity, which was used during the repeated charge/discharge cycles before. Because of this the whole capacity of the NiCd batteries should be used for each discharge cycle to avoid a decrease of the maximum capacity [1]. Another disadvantage is the presence of cadmium, which is an environmental hazard. This may lead to a complete ban on NiCd batteries in the future.

2.2.2. NiMH batteries. The major difference between NiCd and NiMH batteries is the fact that in a NiMH battery a metal hydride alloy is used for the electrode instead of the cadmium. In this way a higher energy density is obtained, the memory effect is reduced and the environmental impact is decreased. Besides, NiMH batteries can replace NiCd batteries because of the same cell voltage of 1.2 V. Applications include cordless and mobile phones, shavers, camcorders, portable audio products and laptop computers. A disadvantage of the NiMH battery is the relatively high self-discharge rate and the relatively low robustness with respect to overcharging, which is made worse by the fact that it is more difficult to detect the battery-full condition during charging [1].

2.2.3. Li-ion batteries. A schematic representation of a typical Li-ion cell is shown in figure 3 [10]. The cell consists of five regions (from left to right in figure 3): the negative electrode current collector made of copper, the porous composite negative insertion electrode, the porous separator, the porous composite positive insertion electrode and the positive electrode current collector made of aluminium. The composite electrodes are made of their active material particles, held together with a binder and a suitable filler material such as carbon black. At the beginning of discharge, the negative electrode is fully lithiated, while the positive electrode is ready to accept lithium ions. During discharge, the lithium ions deintercalate from the negative electrode particles and enter the solution phase, while in the positive electrode region lithium ions in the solution phase intercalate into the $LiCoO_2$ particles. This results in a concentration gradient, which drives lithium ions from the negative electrode to the positive electrode. The cell voltage decreases during discharge, as the equilibrium potentials of the two electrodes are strong functions of the concentrations of lithium on the surface of the electrode particles. The cell is considered to have reached the end of discharge when its voltage drops to 3.0 V [10].

The positive electrode is built of lithium metal oxides (e.g., $LiCoO_2$, $LiNiO_2$ or $LiMn_2O_4$) to store the lithium ions. The negative carbon electrode is made of graphite or petroleum coke. The electrolyte is mostly a salt dissolved in an organic solvent, but batteries with other solvents such as propylene carbonate also exist. An example of an applied salt is $LiPF_6$.

The operating voltage of the Li-ion batteries is critical and over(dis)charging results in fast ageing and may cause fire or even exploding batteries. These require an essential electronic protection circuit to prevent over(dis)charging as discussed in section 1. Applications include mobile phones, shavers, camcorders, portable audio products and laptop computers.

2.3. Summary

Several types of rechargeable battery systems have been discussed in this section. The main characteristics of the discussed battery types are summarized in table 1 [1].

3. History of state-of-charge indication

In this section, previous and current SoC technologies will be presented. Almost as long as rechargeable batteries have



Figure 4. Astronauts exploring the Moon in Lunar Roving Vehicles in 1971–1972 relied on early Curtis gauges [13].

Table 1. Overview of the main characteristics of the most important rechargeable battery systems [1].

Battery system	NiCd	NiMH	Li-ion
Average operating voltage (V)	1.2	1.2	3.6
Energy density (W h l^{-1})	90–150	160–310	200–280
Specific energy (W h kg^{-1})	30–60	50–90	90–115
Self-discharge rate (%/month) at 20 °C	10–20	20–30	5–10
Cycle life	300–700	300–600	500–1000
Temperature range (°C)	−20–50	−20–50	−20–50

existed, systems able to give an indication about the amount of charge that is available inside a battery have been around. In 1963, Curtis Instruments pioneered gauges to monitor the SoC, the 'fuel' level, of vehicle traction batteries. One of the methods used by Curtis involves predicting the remaining capacity of a battery by measuring the elapsed time period since the loaded voltage dropped below a certain value. For example, when the battery is discharged from 24.25 V for a period of 3 min the remaining capacity is lowered from 100% to 90%, etc [11]. A compensation method for different discharge rates is also presented.

Several monitors based on the average voltage were available in that period of time, such as Sears battery monitor, range indicator (by Motovator) and the above-mentioned Curtis fuel gauge. Of these, the Curtis fuel gauge has been found to be the most sophisticated and accurate [12]. Curtis SoC gauges were even used on the Moon (see figure 4) [13]. In one of the Curtis patents filed by Finger *et al* in 1975 the current flowing from the battery is sent to an integrator module, which registers the current depletion [14]. During charging, the current is integrated in the integrator module providing a continuous display of the SoC and information to regulate the charge rate.

An account of attempts that have been made to develop an SoC indicator for a nickel–cadmium battery was given by Lerner in 1970 [15]. He concluded that the only reliable way of estimating the SoC is to use a current-sharing method. In this method, the current output of a battery having a known SoC is compared with that of a battery having an unknown SoC. From this comparison, the SoC of the unknown battery can be deduced.

In 1974, York *et al* introduced an SoC indicator in which the value of the measured battery voltage is indicated with respect to two voltage levels stored in the system [16]. In a first state it is indicated that the battery voltage is greater than a first voltage level and in a second state it is indicated that the battery voltage is less than the first voltage level but greater than a second voltage level. Finally, a third state indicates that the battery voltage is less than the second voltage level and disables the load connected to the battery. The magnitude and duration of the voltage reduction are monitored by a threshold circuit, which produces an output whenever the terminal voltage falls below the lower threshold value. As a response to the voltage reduction, a number of pulses is generated. An electronic counter is used for counting the pulses and accumulating the counts. An integral proportional to the total time that the terminal voltage is below the lower threshold voltage is generated. The output of the integration provides an indication of the SoC. A principal advantage of the presented method is that SoC can be indicated despite sudden disconnection and reconnection of the battery.

The main concept of determining the SoC of a battery based on a comparison between the measured battery voltage and predetermined threshold values that correspond to different SoC values is also presented in [17, 18]. In [17], it is shown that the approximation of the accumulated operating time of a rechargeable battery is a function of the internal resistance of the rechargeable battery. The SoC is measured relative to the maximum capacity of the rechargeable battery. The application is in the field of mobile telephony. In [18], voltage levels are measured during battery charge and discharge and compared with predetermined values, which are modified as a function of temperature. The stored battery charge or discharge curves are divided into curve portions defined by voltage levels and rates of changes of the voltage levels. Each curve portion defines a particular SoC of the battery. Measured voltage levels and rates of changes are then associated with the predetermined charge or discharge curve portion and the defined SoC.

In 1974, Brandwein *et al* developed a device for monitoring nickel–cadmium batteries [19]. In addition to voltage measurements, the current that flows into and out of the battery and the battery temperature are also measured and used in order to provide SoC indication. The battery voltage is stored as an analogue voltage signal at different temperatures. In [20], an equivalent circuit diagram of the battery, which uses as inputs current, voltage and temperature measurements, is presented. The measured values are compared with the corresponding values obtained from calculations with the equivalent circuit diagram. The parameters in the equivalent circuit diagram and the state variables are varied such that the measured values are matched and the SoC is determined from the adjusted parameters.

Christianson *et al* developed a method in which the SoC of a battery is indicated based on the open-circuit voltage (OCV) calculation in 1975 [21]. The OCV is directly proportional to the battery SoC and can be calculated using the following equation:

$$\mathrm{OCV} = V_{\mathrm{term}} + I\,R \tag{1}$$

where V_{term} is the battery terminal voltage, I the actual battery current—considered as a positive value during discharge and as a negative value during charge and R is the internal

resistance. Note that OCV $= V_{\text{term}}$ when $I = 0$, but after current interruption this takes a while due to several relaxation processes occurring inside a battery. In addition to the OCV, the method presented by Eby *et al* in 1978 also uses the voltage under load to determine the SoC of a LA storage battery during a discharge cycle [22]. The battery's initial OCV is stored in a settable memory. It has been demonstrated that in the case of LA batteries the OCV has a linear correlation to the charge level of the battery under a defined set of circumstances. At each moment in time the discharge rate can be determined as a comparison function between voltage under load and the corresponding OCV.

The first impedance measurements of batteries appear to have been made by Willihnganz in 1941 [23]. These involved excitation of the electrochemical cell by an ac voltage of small amplitude of about 5 mV and evaluation of the resistive and the reactive components or other related parameters such as the modulus of impedance and phase angle. As such measurements encompass a wide range of ac signal frequencies, various characteristic parameters of the electrochemical cell and the kinetics of the associated reactions can be evaluated [23]. As an alternative, Dowgiallo *et al* (1975) and Zaugg (1982) developed methods for determining the SoC of a battery based on impedance measurements [24, 25]. The phase angle between the ac voltage across the battery terminals and the ac current through the battery (measured as a voltage through a sense resistor) is continuously monitored. The method from [24] is presented in the field of nickel–cadmium batteries and applicable in equipment such as transmitters, receivers, tape recorders, movie cameras, aircraft, electric vehicles, small calculators and computers. In the system developed by Muramatsu in 1985 the relationships between the battery impedance at different frequencies (defined as impedance spectroscopy), remaining capacity and SoH is used to detect SoC and SoH of the battery [26]. Predetermined values based on this relationship are stored in look-up tables and used in order to determine SoC and SoH of the battery. Look-up tables are tables in which fixed values of measured parameters, such as voltage, current, impedance and temperature, can be stored and used in order to indicate the SoC.

In 1984, Peled developed a method for determining the SoC of lithium-ion batteries [27]. At the base of the presented method are predetermined voltage and temperature measurements, used as input parameters for look-up tables. After applying a current step and a short rest period, the OCV of a battery is measured as a function of temperature. The measured value is compared with a corresponding predetermined value stored in a look-up table. Based on this comparison, the SoC is indicated. In the system of Kopmann (1987) the terminal voltage, the current and time are measured during each battery charging and discharging cycle [28]. These values are used as inputs for look-up tables. The characteristic of the terminal voltage curve during charging and discharging is used for minimizing the differences between the measured values and the actual SoC of the battery. In [29], a method for determining the SoC for NiMH batteries in notebook applications is presented. The method uses the battery's temperature, voltage and discharge/charge rate measurements to determine SoC, using look-up tables. A possible hardware and software implementation is also presented. By modifying the look-up tables, these techniques can be employed for many other battery technologies.

In 1981, Finger of Curtis Instruments patented a method where the SoC of LA batteries is determined during a quiescent interval with no current flow through the battery [30]. After applying a current step, the battery terminal voltage is measured and the combination of these two measurements (battery voltage and time) is used for battery OCV recovery characteristics determination. This predictable time function of voltage recovery is substantially independent of the actual voltage level of the terminal voltage.

In [31], an SoC indicator for a lithium-ion battery based on a cell OCV is presented. The SoC is calculated based on a stored SoC–OCV relationship. The SoC–OCV relationship is obtained by defining the battery charge amount when the OCV of the cell is 3.9 V as SoC $= 100\%$ and defining the battery charge amount when the OCV of the cell is 3.5 V as SoC $= 0\%$. By defining SoC $= 100\%$ and SoC $= 0\%$, the SoC can be correctly calculated and displayed even when the battery ages.

The methods presented in [32–36] use Coulomb counting, i.e., battery current integration, as a basis. The method developed by Aylor (1992) holds for LA batteries [32]. The described technique is a combination of the previously described OCV method and coulometric measurements (Coulomb counting). The paper states that combining these two measurement techniques, it is possible to cover the weakness of both techniques and provide an accurate SoC indication. Coulometric measurements are used in short-time operations, where the accumulation of error is negligible. The error that is accumulated from coulometric measurement techniques can be corrected when an OCV reading is taken every time the battery is sufficiently rested. In order to reduce the required rest period of the OCV measurements, a method has been developed to predict the OCV before the battery voltage is fully stabilized. It should be noted that this method has been limited to LA-type batteries. This restriction is used because of the unique linear relation OCV–specific gravity that exists in LA batteries. In [37], it is indicated that the method presented in [32] could provide 99% accuracy (1% error) for the charge detection but at a high cost for realization. The application described in [35] relates to a battery pack utilizing NiMH batteries or any other battery technology such as LA, lithium polymer, etc, operating in a hybrid–electric power train for a vehicle. In addition to Coulomb counting, the systems presented by Kikuoka [33] and by Seyfang [34] also compensate for temperature, charging efficiency of the battery, self-discharge and ageing. In [34], the capacity of the battery is monitored and compared with the initial capacity. In this way, an indication of the SoH of the battery is provided. When the battery is fully discharged or when the battery is fully charged a particular set of parameters, such as the conversion efficiency of each battery, are 'learned' and updated to take into account the ageing of the battery. In addition to Coulomb counting and voltage, current and temperature measurements, a mathematical model that is implemented on a computer that simulates the behaviour of the battery is also presented in [36].

The methods presented in [38–40] also use adaptive methods for determining the SoC of a battery. In 1997, Gerard *et al* developed a method in which the 'state variables' of the

battery are replaced with neural weights, aiming at providing the user of portable equipment with an accurate estimation of the remaining working time, i.e., how much time is left until the battery voltage reaches the end-of-discharge voltage defined in a portable device [38]. Two artificial neural networks are used for modelling the system implementation, more precisely for adapting the prediction of the current discharge curve to the general behaviour of the used battery pack. A mean error of about 3% has been found using this implementation method. In 1999, Salkind *et al* developed a method for the SoC and SoH prediction, based on fuzzy logic modelling, for two battery systems, lithium–sulphur dioxide (Li–SO_2) and NiMH [39]. The method involves the use of fuzzy logic mathematics to analyse data obtained by impedance spectroscopy and/or Coulomb counting techniques. Using the above method for a limited data set, the obtained maximum error between the measured SoC and the model-predicted SoC in the case of lithium–sulphur dioxide cells was ±5%. In 2000, Garche *et al* developed a method in which the Kalman filters (KF) are presented for implementing an adaptive method in connection with parameter estimation to determine SoC [40]. The basis of the filter is a numeric battery model description. The battery voltage is estimated on the basis of the current and temperature measurements and then the results are compared with the measured value of the battery voltage. Adaptivity of the model is based on a comparison of the estimated values with observed battery behaviour. A more detailed description of these adaptive methods will be given in section 5.

In 2000, Bergveld *et al* developed a method of estimating the SoC of a rechargeable lithium battery [1, 41]. The basis of the algorithm is current measurement during the charge or discharge state and voltage measurement during the equilibrium state (state in which no current is flowing into or out of the battery and all the conditions inside the battery are fully stabilized). In the charge and discharge states, the determination of the SoC relies on calculating the charge withdrawn from or supplied to the battery by means of current integration and subtracting this charge from or adding it to the SoC calculated earlier. Hence, in these states Coulomb counting is applied, where the battery is viewed as a simple linear capacitor.

In the discharge state, apart from simple Coulomb counting the effect of the overpotential is also considered. Due to this overpotential, the battery voltage during discharging is lower than the electro-motive-force (EMF equals the sum of the equilibrium potentials of the electrodes of the battery), which in equilibrium equals the OCV presented earlier. The value of the overpotential depends on the discharge current, the SoC, age and temperature. Especially at low temperatures and at low SoC, due to a high overpotential mainly caused by diffusion limitation of the electrochemical reactions, the remaining charge cannot be withdrawn from the battery, because the battery voltage will drop below the end-of-discharge voltage defined in the portable device. This leads to an apparent capacity loss, which at low temperatures of, e.g., 0 °C can amount to more than 5%. Hence, a distinction should be made between available charge in the battery and the charge that can be withdrawn from the battery under certain conditions. As overpotentials are temperature dependent, temperature measurements are also needed in the discharge state [1].

Table 2. History of SoC development.

Year	Researcher/ company	Method
1963	Curtis	Voltage measurements
1970	Lerner	Comparison between two batteries (one has a known SoC)
1974	York	Threshold in voltage levels
1974	Brandwein	Voltage, temperature and current measurements
1975	Christianson	OCV
1975	Dowgiallo	Impedance measurements
1975	Finger	Coulomb counting
1978	Eby	OCV and voltage under load
1980	Kikuoka	Book-keeping
1981	Finger	Voltage relaxation
1984	Peled	Look-up tables based on OCV and T measurements
1985	Muramatsu	Impedance spectroscopy
1986	Kopmann	Look-up tables based on V, I and T measurements
1988	Seyfang	Book-keeping and adaptive system
1992	Aylor	OCV, OCV prediction and coulometric measurements
1997	Gerard	Voltage and current measurements, artificial neural networks
1999	Salkind	Coulomb counting, impedance spectroscopy, fuzzy logic
2000	Garche	Voltage and current measurements, Kalman filters
2000	Bergveld	Book-keeping, overpotential, EMF, maximum capacity learning algorithm

In the equilibrium state, the battery's SoC is determined by means of voltage measurement. Because only a small current flows in this state, the measured voltage approaches the EMF of the battery. The algorithm uses a stored EMF-versus-SoC curve to translate a measured voltage value into an SoC value expressed in % of the maximum capacity. The EMF-versus-SoC curve remains the same even when the battery ages and additionally, the temperature dependence of this curve is relatively low [1, 42]. The EMF method also works as a calibration of the SoC system, because for a certain measured EMF, the same SoC is found irrespective of the age and temperature of the battery. This calibration is important, because in charge and discharge states, the calculated SoC will eventually drift away from the real value due to, e.g., measurement inaccuracy in the current and the integration in time of this inaccuracy [1].

Table 2 summarizes the most important points of the presented history in SoC development. As will be shown later in this review, the actual state-of-charge indication integrated circuits (IC) are based on the methods described in this table. More detailed information on the applied methods will be given in section 5.

4. A general state-of-charge system

In science, the standard unit for battery capacity is the coulomb (named after the French physicist C A Coulomb, 1736–1806), which describes the time the battery can produce a given current. The coulomb is the unit of electric charge equal to one ampere-second (A s). In practice, however, cell or battery capacity is more commonly expressed in

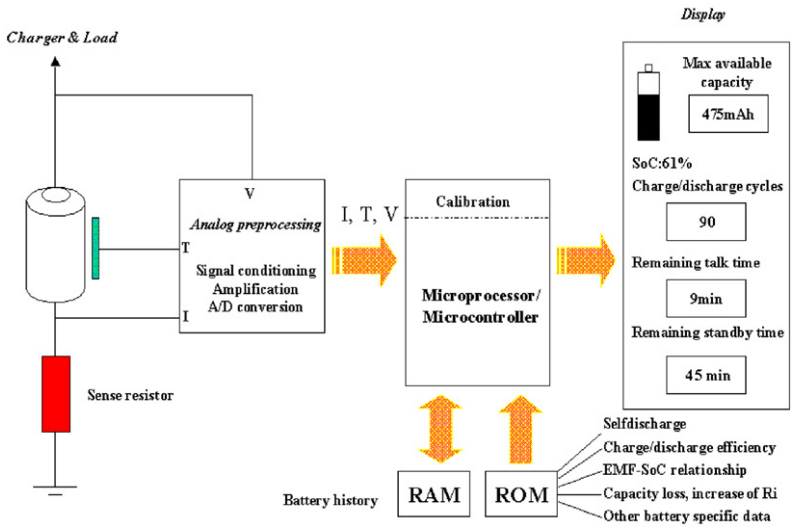


Figure 5. General functional architecture of a state-of-charge system.

ampere-hours (A h) or milliampere-hours (mA h). One of the most relevant characteristics for a user is to know the SoC of the battery. In [35] SoC is defined as the percentage of the full capacity of a battery that is still available for further discharge or in [23] as the ratio of the available capacity of a cell to its maximum attainable capacity. In order to understand what the term 'SoC' really represents, a clear definition must be given: SoC is the percentage of maximum possible charge that is present inside the battery. The SoC measurement method and the computational model based on the correct SoC definition must be simple, convenient, practical and reliable.

Figure 5 shows an example of a practical SoC system. The battery may include a plurality of battery cells connected in series, each of the battery cells having at least two terminals. The SoC system may include an analogue-to-digital converter (ADC) for converting a voltage drop between at least two sense resistor connection pins as a measure of the current (I) into a digital signal and also for converting the measured analogue values of the battery voltage (V) and temperature (T) into digital signals.

A microprocessor/microcontroller (in which the SoC algorithm is stored) determines the SoC of the battery system based on the measured signals. Two types of memory are needed. Basic battery data, such as the amount of self-discharge as a function of T and the discharging efficiency as a function of I and T, are read from the read-only memory (ROM). When the SoC algorithm is based on EMF measurements, the EMF–SoC relationship can be stored in ROM together with other battery specific data. The random access memory (RAM) is used to store the history of use, such as the number of charge/discharge cycles, which can be used to update the maximum battery capacity. Each part of this system (software algorithm or hardware device) will influence the final accuracy in the SoC indication (e.g., inaccuracy in the V, T and I measurements will give an inaccuracy in the final SoC). An important role must be accorded also to the calibration of the SoC, because, for example, if the SoC algorithm is based on current measurement and integration the obtained error caused by the current measurement inaccuracy will accumulate over time.

5. Possible state-of-charge indication methods

In order to efficiently discuss SoC indication methods, some of the common terms used in the battery SoC industry have to be defined. Below some battery-related definitions concerning the SoC application are given [1]:

Ampere-hour. A measure for electric charge defined as the integral product of current (in Amperes) and time (in hours).

Cell. The basic electrochemical unit used to generate electrical energy from stored chemical energy or to store electrical energy in the form of chemical energy. A cell consists of two electrodes in a container filled with an electrolyte.

Battery. Two or more cells connected in an appropriate series/parallel arrangement to obtain the required operating voltage and capacity for a certain load. The term battery is also frequently used for single cells.

C rate. A charge or discharge current equal in Amperes to the rated capacity in A h. Multiples larger or smaller than the C rate are used to express larger or smaller currents. For example, the C rate is 1100 mA in the case of an 1100 mA h battery, whereas the $C/2$ and $2C$ rates are 550 mA and 2.2 A, respectively.

Cycle life. The number of cycles that a cell or battery can be charged and discharged under specific conditions, before the available capacity in A h fails to meet specific performance criteria. This will usually be 80% of the rated capacity.

Cut-off voltage. The lowest operating voltage of a cell at which the cell is considered depleted. Also often referred to as end-of-discharge voltage or final voltage [32].

State-of-charge (SoC). The percentage of the maximum possible charge that is present inside the battery.

State-of-health (SoH). A 'measurement' that reflects the general condition of a battery and its ability to deliver the specified performance compared to a fresh battery. It takes into account such factors as charge acceptance, internal resistance, voltage and self-discharge rate.

Remaining run-time. The estimated time that the battery can supply charge to a portable device under valid discharge conditions before it will stop functioning. The remaining run-time can be inferred from the remaining capacity in two ways, depending on the type of load [1]: for a current-type load the remaining capacity in mA h, so expressed as charge, is divided by the drawn current in mA and for a power-type load the remaining capacity in mW h, so expressed as energy, is divided by the drawn power in mW.

Accuracy. The qualification of the expected closeness of the result of a measurement and the true value. The generally accepted quantitative measure related to accuracy is uncertainty. The uncertainty of a measurement is a parameter that characterizes the range of the values within which the true value of a measurement lies. It expresses the lack of knowledge that we have on the measurand (the particular quantity to be measured) [43]. For SoC indication, for example in the form of a remaining run-time indication, an uncertainty less than a few per cent is needed under all practical conditions, including ageing of the battery, to yield a reliable system in the perception of the user.

As already described in section 3, several methods in the art of determining the SoC of a cell or battery of cells are known. Some older, very inexpensive fuel gauges simply measured voltage. This battery voltage is a highly inaccurate indication of a battery's capacity, because it changes with temperature, discharge rates and ageing. Another known method of measuring SoC involves impedance measurements utilizing bridge circuits that are manually adjusted. The measurements obtained are compared with previously generated standard reference curves. Still another prior-art method used to determine the battery SoC involves estimating the SoC by means of the battery response to current or voltage pulses. These pulse systems provide a rough means of determining the SoC and are used primarily to determine if a battery is still useful. This first group of methods will be called *direct measurements*.

Another known method is to measure the current flowing into and out of a battery and to integrate this current over time in order to determine its capacity [1]. When utilizing these current integrators one must correct the estimation of the SoC obtained, since several battery-related factors affect the accuracy of this estimation. These factors include temperature, history, charge and discharge efficiencies and cycle life. The integration of current is referred to in literature as *Coulomb counting* [1]. When this method is compensated for discharging 'efficiency', self-discharge and capacity loss then this method is presented as a *book-keeping* system [1].

The main problem in designing an accurate SoC indication system is the unpredictability of both battery and user behaviour. In this case an adaptive system has to be used, which is based on direct measurement, book-keeping or a combination of the two [1]. In order to make all the aspects more clear, each of these methods will be presented below in detail.

5.1. Direct measurement

The direct measurement method refers to the measurement of battery variables such as the battery voltage (V), battery impedance (Z) and voltage relaxation time (τ) after application of a current step. Most relations between battery variables and the SoC depend on the temperature (T). Therefore, besides the voltage or impedance, the battery temperature should also be measured. The basic principle of an SoC indication system based on direct measurement is shown in figure 6.

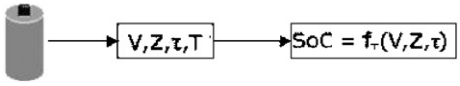


Figure 6. Basic principle of an SoC indication system based on direct measurement.

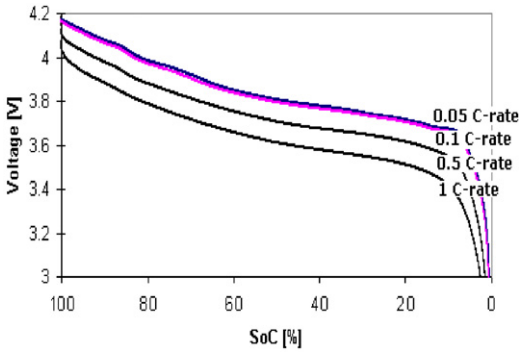


Figure 7. Li-ion cell's voltage curve at different discharge rates.

The main advantage of a system based on direct measurement is that it does not have to be continuously connected to the battery. The measurements can be performed as soon as the battery has been connected [1].

5.1.1. Voltage measurements. Although the use of voltage measurement has been a popular method, especially for mobile phone applications, it does not produce the most accurate results available. Determining the remaining capacity of a cell by simply measuring its voltage level may be less expensive and may use less computing power of the host CPU compared to Coulomb counting, but under real life conditions voltage measurements alone can be very misleading [44]. While it is true that a given cell voltage level will continually drop during discharge, the voltage level relationship to remaining charge varies greatly over cell temperature and discharge rate. Figure 7 shows a Li-ion cell's voltage curve during discharge at different discharge rates.

The important point illustrated in figure 7 is the relationship between the cell voltage and its discharged capacity. It can be seen that the voltage discharge curve strongly depends on the discharge rate. The error in SoC estimation based on voltage measurement can be corrected by the system if the dependence of the battery voltage on the cell temperature and discharge rate is known. However, if those measured curves are included, the process becomes more complicated and expensive than a Coulomb counting approach [44].

5.1.2. The EMF method. The term EMF stands for electromotive force. It is the internal driving force of a battery for providing energy to a load. In principle, the EMF can be calculated from thermodynamic data and the Nernst equation

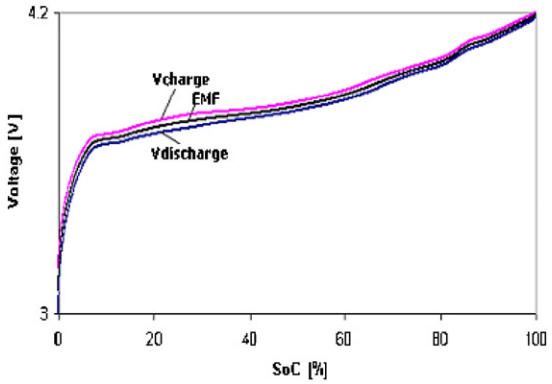


Figure 8. EMF curve obtained with the linear interpolation method.

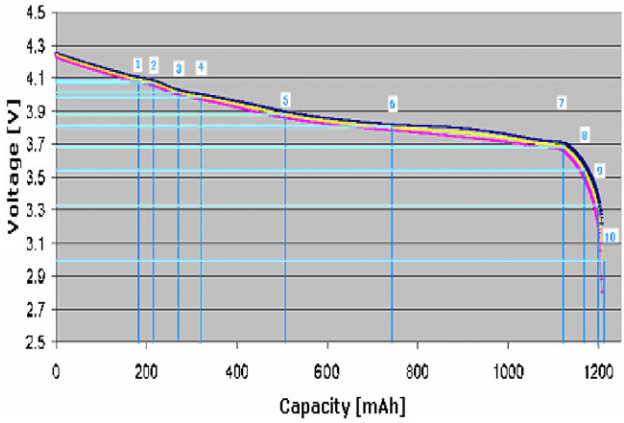


Figure 9. Piecewise linear EMF curve for US18500G3 Li-ion battery (Sony).

Table 3. Possible EMF curve implementation (see also figure 9).

Interval number	Interval voltage limits (V)	SoC (%)
1	4.08–4.24	85–100
2	4.06–4.08	81.7–85
3	4.02–4.06	76.7–81.7
4	3.98–4.02	73.4–76.7
5	3.88–3.98	58.4–73.4
6	3.8–3.88	22–58.4
7	3.68–3.8	8.7–22
8	3.54–3.68	5.4–8.7
9	3.32–3.54	2.1–5.4
10	3–3.32	0.5–2.1
11	2.5–3	0–0.5

(or equations derived thereof) [1]. Another method with which the EMF can be obtained is called *linear interpolation*. In this method the average battery voltage, calculated at the same SoC, is determined from the battery voltages during two consecutive discharge and charge cycles with the same currents and at the same temperature. Figure 8 shows the EMF curve obtained at 25 °C with the linear interpolation method in the case of the US18500G3 Li-ion battery (Sony).

In another known method, the EMF is determined based on *voltage relaxation*. The battery voltage will relax to the EMF value after current interruption. This may take a long time, especially when a battery is almost empty, at low temperatures and after a high discharge current rate [42]. In another EMF determination method—*linear extrapolation*—the battery voltages obtained with different currents with the same sign and at the same SoC value are linearly extrapolated to a current of value zero [1].

The EMF of a Li-ion battery is observed to be a good measure for battery SoC. It has been demonstrated that the relationship between EMF and SoC does not change during cycling of the battery, if SoC is expressed in relative capacity [1]. Temperature dependence of the EMF is small, except when the battery is near to fully discharged or near to fully charged.

When the SoC algorithm is based on the EMF an accurate method for EMF implementation is required. Three of the practical EMF implementation methods will be presented below.

(a) *Look-up table.* A table in which fixed values of the measured parameters can be stored and used in order to indicate SoC. The size and the accuracy of the look-up tables in SoC indication systems depend on the number of stored values. One of the main drawbacks of this method is that even in the case of a single battery type it is impossible to take into account every point of the EMF curve in order to provide an accurate SoC indication system. Even if many measurement points are included, the process becomes more complicated and expensive than other approaches and probably does not provide any significant advantages.

(b) *Piecewise linear function.* In this method the EMF curve is approximated with piecewise linear functions. A possible example with 11 intervals is shown in figure 9 for the US18500G3 Li-ion battery (Sony). The intervals in voltage and the corresponding SoC are presented in table 3.

Based on equation (5.1), the SoC can be calculated for any measured battery equilibrium voltage value V_m:

$$\mathrm{SoC} = \mathrm{SoC}_{\mathrm{low}} + \frac{V_m - V_{\mathrm{low}}}{V_{\mathrm{high}} - V_{\mathrm{low}}}(\mathrm{SoC}_{\mathrm{high}} - \mathrm{SoC}_{\mathrm{low}}) \qquad (5.1)$$

where V_{low} and V_{high} are fixed and specific values from the EMF curve for the voltages corresponding to the $\mathrm{SoC}_{\mathrm{low}}$ (%) and $\mathrm{SoC}_{\mathrm{high}}$ (%) SoC values, like e.g. in table 3 $V_{\mathrm{low}} = 4.08$ V and $V_{\mathrm{high}} = 4.24$ V corresponding to $\mathrm{SoC}_{\mathrm{low}} = 85\%$ and $\mathrm{SoC}_{\mathrm{high}} = 100\%$, respectively.

When enough voltage and SoC intervals are chosen this method will give more flexibility (possibility to be implemented for other battery types) and precision in SoC estimation based on the EMF curve when compared to a look-up table implementation. The problems of spread, temperature and ageing remain to be solved.

(c) *Mathematical function.* In this method, the EMF curve is approximated with a mathematical function. A possible example will be further described. The EMF of a Li-ion battery with intercalated electrodes is modelled as a difference in equilibrium potentials of positive and negative electrodes [1], correspondingly, i.e.,

$$U_{\mathrm{emf}} = E^{\mathrm{eq}}_{\mathrm{LiCoO_2}} - E^{\mathrm{eq}}_{\mathrm{LiC_6}} \qquad (5.2)$$

where equilibrium potentials of the positive and negative electrodes are modelled as follows. The equilibrium potential

of the positive electrode is given by

$$E^{\mathrm{eq}}_{\mathrm{LiCoO_2}} = E^{0}_{\mathrm{LiCoO_2}} - J\left(\log\left(\frac{x_{\mathrm{Li}}}{1-x_{\mathrm{Li}}}\right) + U^{+}_{j}x_{\mathrm{Li}} - \zeta^{+}_{j}\right) \tag{5.3}$$

$$\zeta^{+}_{2} = \left(U^{+}_{2} - U^{+}_{1}\right)^{*}x_{\mathrm{ph}} + \zeta^{+}_{1}, \qquad j = \begin{cases} 1 & x_{\mathrm{ph}} \leqslant x_{\mathrm{Li}} \leqslant 1 \\ 2 & 1/2 \leqslant x_{\mathrm{Li}} \leqslant x_{\mathrm{ph}} \end{cases} \tag{5.4}$$

where $E^{0}_{\mathrm{LiCoO_2}}$ is the standard redox potential of the $LiCoO_2$ electrode in V, U^{+}_{j} denotes the dimensionless interaction energy coefficient in the $LiCoO_2$ electrode, ζ^{+}_{j} denotes a dimensionless constant in the $LiCoO_2$ electrode, x_{Li} is the mol fraction of Li^+ ions inside the positive electrode, i.e., the SoC of the $LiCoO_2$ electrode, and $J = RT/F$, where R is the gas constant (8.314 J (mol K)$^{-1}$), F is Faraday's constant (96 485 C mol^{-1}) and T is the (ambient) temperature in K. In equation (5.4), a phase transition occurs at $x_{\mathrm{Li}} = x_{\mathrm{ph}}$, that results in a curvature change. According to modern literature on Li-ion with a $LiCoO_2$ electrode (see, e.g., [1, 46]), the main phase-transition point is located nearby $x_{\mathrm{ph}} \approx 0.75$, while there is also a minor one at $x_{\mathrm{ph}} \approx 0.93$. The negative electrode is modelled symmetrically, with only a small difference.

$$E^{\mathrm{eq}}_{\mathrm{LiC_6}} = E^{0}_{\mathrm{LiC_6}} - J\left(\log\left(\frac{z_{\mathrm{Li}}}{1-z_{\mathrm{Li}}}\right) + U^{-}_{j}z_{\mathrm{Li}} - \zeta^{-}_{j}\right) \tag{5.5}$$

$$\zeta^{-}_{2} = (U^{-}_{2} - U^{-}_{1})^{*}z_{\mathrm{ph}} + \zeta^{-}_{1}, \qquad j = \begin{cases} 1 & 0 \leqslant z_{\mathrm{Li}} \leqslant z_{\mathrm{ph}} \\ 2 & z_{\mathrm{ph}} \leqslant z_{\mathrm{Li}} \leqslant 1 \end{cases} \tag{5.6}$$

where $E^{0}_{\mathrm{LiC_6}}$ is the standard redox potential of the LiC_6 electrode in V, U^{-}_{j} denotes the dimensionless interaction energy coefficient in the LiC_6 electrode, ζ^{-}_{j} denotes a dimensionless constant in the LiC_6 electrode, z_{Li} is the mol fraction of the Li^+ ions inside the negative electrode, i.e., SoC of the negative electrode. In the negative electrode a phase transition occurring around $z_{\mathrm{ph}} \approx 0.25$ is modelled. Under normal operational conditions x_{Li} will cycle between 0.5 and 1, but z_{Li} will cycle between 0 and 1 [1]. Using an adaptive method for updating the equation parameters taking into consideration factors like spread, temperature, ageing of the batteries, this last presented method will probably offer the best solution for a practical EMF implementation.

5.1.3. Impedance measurements. The ratio of a complex voltage and a complex current is in general a complex quantity. The ratio V/I is generally denoted as the impedance Z (see for instance [47]). This definition is not always correctly applied in the battery-related literature. Impedance data on many practical battery designs have been reported in [48]. A useful means of studying processes in electrochemical systems including biological processes, batteries and capacitors is to make impedance measurements over a wide range of frequencies, usually referred to as electrochemical impedance spectroscopy (EIS).

The electrochemical impedance (or ac impedance) of a battery characterizes its dynamic behaviour, that is, its response to an excitation of small amplitude. In principle, any type of excitation signal may be used (sine wave, noise, step, ...). However, in practice, sine waves are employed most of the time. In galvanostatic (constant current) mode, the dc current I (polarization current) charging or discharging the battery is modified using a sinusoidal current:

$$\Delta I = I_{\mathrm{max}} \sin(2\pi f t) \tag{5.7}$$

at frequency f, which is superimposed onto I, yielding a sinusoidal voltage response

$$\Delta V = V_{\mathrm{max}} \sin(2\pi f t + \phi) \tag{5.8}$$

around the dc voltage V at the terminals of the battery. The amplitude V_{max} and the phase angle ϕ depend on the frequency f and V_{max} also depends on the amplitude I_{max} of the applied ac current. In contrast, in potentiostatic mode (constant voltage), the dc voltage V at the terminals of the battery is modified using a sinusoidal voltage

$$\Delta V = V_{\mathrm{max}} \sin(2\pi f t) \tag{5.9}$$

at frequency f, which is superimposed onto V, yielding a sinusoidal current response

$$\Delta I = I_{\mathrm{max}} \sin(2\pi f t - \phi) \tag{5.10}$$

around the dc current I flowing through the battery. In that case, the amplitude I_{max} and the phase angle ϕ depend on the frequency f and I_{max} also depends on the amplitude V_{max} of the applied ac voltage. In both cases, the impedance is defined by

$$Z(f) = \frac{V_{\mathrm{max}}}{I_{\mathrm{max}}}\,\mathrm{e}^{\mathrm{j}\varphi}. \tag{5.11}$$

Therefore, the electrochemical impedance of a battery is a frequency-dependent complex number characterized either by its real and imaginary parts or by its modulus and phase angle ϕ.

It must be noted that the voltage amplitude V_{max} must not exceed about 10 mV to ensure that impedance measurements are performed under linear conditions. In that case, the excitation and response signals are actually sine waves, and the measured impedance does not depend on the amplitude of the excitation signal. Such a condition is easily fulfilled in potentiostatic mode where V_{max} is directly imposed by the experimenter. In galvanostatic mode, I_{max} must be determined so that V_{max} is close to 10 mV at all frequencies, especially at the lowest analysed frequency where the modulus of the battery impedance is maximum. High power ac currents (several A) may be required for high capacity batteries whose impedance values are in the mΩ range. Impedance diagrams may be presented as a Bode plot (modulus in log scale versus frequency and phase angle versus frequency) or, more frequently, as a Nyquist plot (imaginary part versus real part). In the latter case, electrochemists generally plot the negative of the imaginary part on the ordinate axis, so that the capacitive loops appear in the upper quadrants. The general shape of the Nyquist diagram of the complex electrochemical impedance of a large capacity LA battery cell is given in figure 10.

The Nyquist diagram of figure 10 presents

- an inductive part at frequencies higher than 100 Hz;
- a high-frequency resistance R_{HF} in the range of mΩ, which is the real part of the impedance at frequencies higher than 100 Hz;
- a first and small capacitive loop (size R_1) for frequencies between 0.1 and 100 Hz corresponding to the electrochemical reaction with the fastest kinetics;

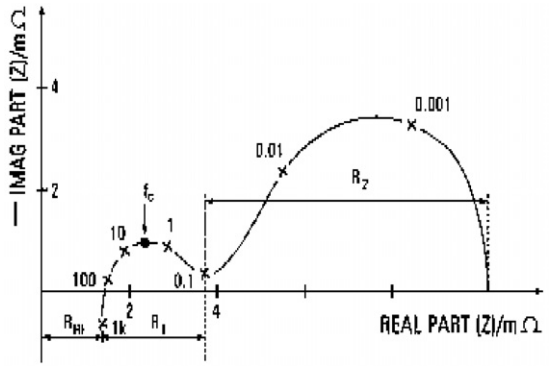


Figure 10. Shape of the Nyquist diagram of the complex impedance of a large capacity lead–acid battery cell (frequencies in Hz) [48].

- a second and large loop (size R_2) for frequencies lower than 0.1 Hz corresponding to the electrochemical reaction with the slowest kinetics.

Measurements of battery impedance as a function of frequency are not practical for SoC indication in a portable device, because a signal with a frequency sweep has to be applied. Some dependence of the impedance on the SoC can be found in a laboratory set-up, but this dependence is usually smaller than the dependence on temperature. The impedance measurements are used in portable products mainly as a means of indicating the battery's condition (SoH). Battery wear-out can be detected by an increase in the internal resistance and therefore assessing the value of the internal resistance by simply applying a current step can serve to test whether the battery is of poor quality and should be replaced [1].

5.2. Book–keeping systems

Book-keeping is a method for SoC indication that is based on current measurement and integration. This can be denoted as Coulomb counting, which literally means 'counting the charge flowing into or out of the battery'. These Coulomb counting data of the battery and other relevant data of the battery such as self-discharge rate of the battery, temperature, charge/discharge efficiency, history (i.e., cycle life), etc will be used as input for the book-keeping system. The following Li-ion battery effects can be compensated for in a book-keeping system [1]:

Discharging 'efficiency'. Depending mainly on the SoC, T and I, only part of the available charge inside a battery can be retrieved. The main mechanisms behind this 'efficiency' are reaction kinetics and diffusion processes. These mechanisms involve reaction-rate and diffusion constants, which are temperature dependent. Moreover, e.g., increased depletion of reacting species at the electrode surfaces occurs at larger currents and reaction-rate constants change over time during ageing of the battery. As a result, a battery that may seem empty after it has been discharged with a relatively high current can still be discharged further after a rest period and/or with a lower current. In general, less charge can be obtained from a battery at low temperatures and/or large discharge currents. A battery's age also influences the discharging efficiency, for example due to increased internal resistance of the battery.

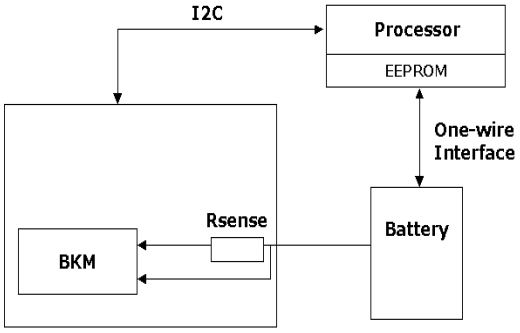


Figure 11. Book-keeping system diagram in a mobile phone application.

Self-discharge. Any battery will gradually lose charge, which becomes apparent when the battery is left unused for some time. A Coulomb counter cannot measure this quantity of charge as no net current flows through the battery terminals. The self-discharge rate of a battery depends strongly on temperature, as well as on the SoC.

Capacity loss. The maximum possible battery capacity in A h decreases when a battery ages. Capacity loss depends on many factors. In general, the more the battery is misused, for example overcharged and overdischarged on a regular basis, the larger the loss will be. Voltage measurement is used to update the maximum battery capacity in most commercial book-keeping systems to deal with capacity loss [1].

The overall accuracy of 'Coulomb counting' depends on the accuracy of the current measurement across the full operating range of the battery system, in both the charging and discharging modes. Typically, the current measuring device measures the voltage across a shunt resistor connected in series with the battery system and converts the measured voltage to a current. This current is integrated and used to determine the SoC of the battery system. The larger current levels require substantially lower shunt resistor values and higher power dissipation ratings. The low resistance of such a shunt results in a very small voltage drop across the shunt that must be measured in order to determine the charge and discharge currents in the battery system. Since the function of the battery monitoring system is to provide time integration of the battery current in order to track the battery SoC, even small errors in the measurement of the current can cause large errors in the SoC measurement to accumulate over time. One of the common errors when the signals to be measured are really small is the offset of the current measurement device.

A particular example of a book-keeping system for a mobile phone application is illustrated in figure 11. The book-keeping module (BKM) continuously monitors the battery and reports the obtained information (voltage, temperature, current measurements and integration) to the processor. The processor uses this information and the battery identification data to determine the SoC. The battery identification data consist of data allowing the determination of the capacity of the battery. Those data are stored in the electrically erasable ROM (EEPROM) and continuously updated by the processor. The telephone uses one-wire interface to communicate with the battery. This means that the battery pack needs only

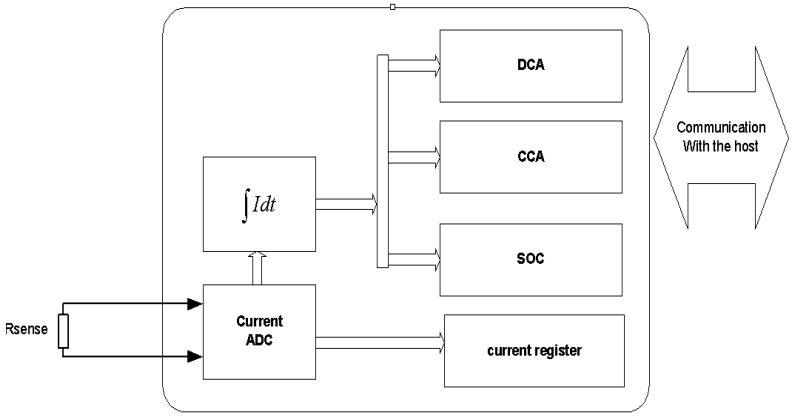


Figure 12. Block diagram of the book-keeping support module (BKM).

three output connections: battery power, ground and one-wire interface.

The BKM can work in two different modes:

The sensitive mode. When the telephone is in idle mode, the consumption of the telephone is low so more accuracy to measure the current is needed. The sensitive mode requires a measurement with high sensitivity.

The normal mode. During communication the consumption current is quite high compared with the idle mode. The normal mode is also used in charge mode.

A high sensitivity, referred to as the lowest current value that needs to be measured, is very important to be sure that all the charge flowing from and to the battery is monitored. The low charge variations are not important for the user, while they are essential for the system reliability. The system must have some internal registers (see figure 12), which accumulate the low variations. These registers are described below.

Current register. Measurement of the current flowing into and out of the battery at the sampling moment.

SOC (SoC counter). It maintains a net accumulated charge flowing into and out of the battery. The reading in this register is an indication of the remaining capacity. This register is reset each time the battery low indication is given.

CCA (charging current accumulator). it accumulates the total charging current over the battery life. It is only updated when the battery receives current.

DCA (discharging current accumulator). It accumulates the total discharging current over the battery life. It is only updated when the battery provides current.

The DCA and CCA registers give information to the battery system needed to determine the end of life of the rechargeable battery, based on total charge/discharge current over its lifetime.

5.3. Adaptive systems

The main problem in designing an accurate SoC indication is the unpredictability of both battery and user behaviour. In this case an adaptive system has to be used, which is based on direct measurement, book-keeping or a combination of the two [1]. Some examples of existing adaptive SoC systems will be described in this section.

In [40, 49–51], optimum Kalman filters are presented for implementing an adaptive method in connection with parameter estimation to determine SoC. The basis of the filter is a numeric battery model description. In [40], the battery voltage is estimated on the basis of the current and temperature measurements and then the results are compared with the measured value of the battery voltage (see figure 13). The inner (internal) parameter contains at least the SoC, but could also contain additional battery variables, such as an estimated value of the battery series resistance (which will give information about SoH of the battery). The model may contain the direct measurement function or the book-keeping function or a combination of the two. The system starts with a basic set of information, which describes standard battery behaviour for the type of battery concerned.

Adaptivity of the model is based on a comparison of the estimated values with observed battery behaviour. This comparison is made whenever possible. The purpose of a Kalman filter is to estimate the state of a system from measurements, which contain errors. The filter has the advantage of being sequential—it needs only the system variables of the previous sample and the forcing terms and observations of the current sample.

As an extension Plett shows in [49–51] how the EKF (extended KF) may be used to adaptively identify unknown parameters in a cell model, in real-time, given cell voltage, current and temperature measurements. Five mathematical state-space models for the purpose of modelling LiPB hybrid electrical vehicle (HEV) cell dynamics are discussed. The models with a single state are simple, but perform the poorest. Adding hysteresis and filter states to the model aids performance, at some cost in complexity. The final presented model includes terms that describe the dynamic contributions due to open-circuit voltage, polarization time constants, electrochemical hysteresis, ohmic loss and the effects of temperature. The results demonstrate that it is possible to achieve root-mean-squared modelling error smaller than the level of quantization error expected in an implementation. It is concluded that EKF provides the best solution for the long-term SoC estimation [51].

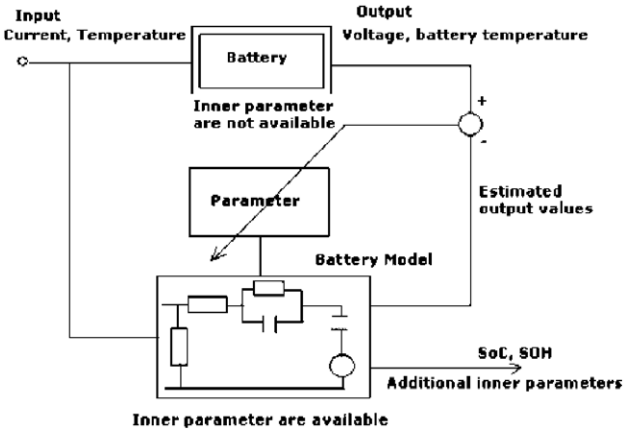


Figure 13. Method for SoC and SoH determination using a Kalman filter [40].

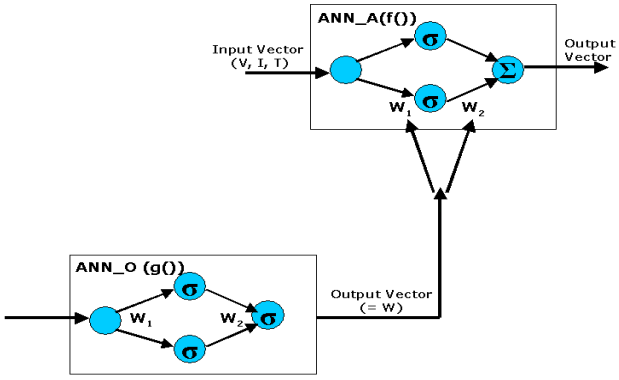


Figure 14. Schematic representation of smart battery management using ANN [38].

In [38] an application is presented, in which the 'state variables' of the battery are replaced with neural weights, aiming at providing the user of portable equipment with an accurate estimation of the remaining working time, i.e., how much time is left until the battery voltage reaches the cut-off value.

Two artificial neural networks (ANN), ANN_A and ANN_O, are used for modelling the system implementation, more precisely for adapting the prediction of the current discharge curve to the general behaviour of the used battery pack (see figure 14). An ANN requires at least two phases: a training phase to set the synaptic weights at those offering the 'best compromise' and an evaluation phase to test the accuracy of the ANN using previously unseen samples. 2860 discharge curves (260 cycles for 11 batteries) have been used to train the system. It has been demonstrated that the ANN (even with online adaptation) can be useful even for small-size products. A mean error of about 3% has been found using this implementation method.

In [52], a method of SoC determination suited for mobile communication application is presented. The effects of pulse-current loads are investigated using a three-layer feed-forward artificial neural network, which is trained using the back propagation algorithm (for adjusting the weights and biases of each neuron based on the error between SoC and the output of the network). The paper demonstrates the use of artificial

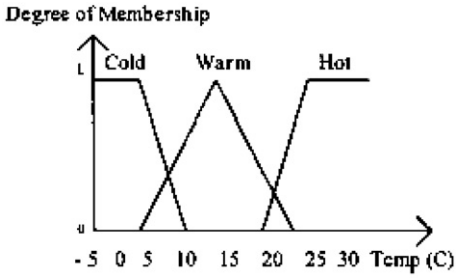


Figure 15. Membership function for temperature.

neural networks for characterizing the discharge patterns for Li-ion batteries under pulsed load typical of those required by the mobile telecommunications system.

In [39], an SoC and SoH prediction, based on fuzzy logic modelling, has been demonstrated for two battery systems, lithium–sulphur dioxide and NiMH. The method involves the use of fuzzy logic mathematics to analyse data obtained by impedance spectroscopy and/or Coulomb counting techniques. Data may be categorized by 'crisp' or 'fuzzy' sets. Crisp sets categorize data with certainty, e.g., a set of temperatures between 30 °C and 40 °C. With fuzzy sets, a set in which data can be categorized is uncertain, e.g., the temperature is 'warm'. This linguistic descriptor 'warm' is a subset of a set of all temperatures and is defined by its membership function. The degree to which an element of the set 'temperature' belongs to the fuzzy subset 'warm' is indicated by a quantity referred to as its 'degree of membership' or fit fuzzy unit value.

Figure 15 shows an example of three subsets, defined by their membership functions, 'cold', 'warm' and 'hot', of the 'universe of discourse' set 'temperature'. Using the above method for a limited data set the obtained maximum error between the measured SoC and the model-predicted SoC in the case of lithium–sulphur dioxide cells was ±5%.

5.4. Summary

A number of possible methods for SoC determination have been presented in this section. Also three of the best-known adaptive methods, which have as inputs battery measured variables such as voltage, impedance, current and use these variables in order to accurately predict the SoC and the remaining time of use for an application, have been described. The SoC determination methods with the application field, advantages and drawbacks are summarized in table 4 [53].

6. Commercial state-of-charge systems

The actual state-of-charge technologies that the main companies use in practice will be presented in this section. Among the battery-management IC producers can be classified: Benchmarq/Unitrode/Texas Instruments, Dallas Semiconductor/Maxim, Linear Technology, PowerSmart/Microchip, Analog Devices, Xicor, Mitsumi and many others (see table 5). Numerous manufacturers are offering SBS (smart battery system)-compatible [54] battery ICs, chargers and software drivers. These companies include AMI, Award Software, Hitachi, MCC, Microchip, O2 Micro, Phoenix Technologies, SystemSoft and VLSI Technologies.

Table 4. Overview of methods for SoC determination [53].

Technique	Application field	Advantages	Drawbacks
Discharge test	Used for capacity determination at the beginning of life	Easy and accurate; independent of SoH	Offline, time intensive, modifies the battery state, loss of energy
Coulomb counting	All battery systems, most applications	Accurate if enough re-calibration points are available and with good current measurements	Sensitive to parasite reactions; needs regular re-calibration points
OCV	Lead, lithium, Zn/Br	Online, cheap, OCV prediction	Needs long rest time (current = 0)
EMF	Lead, lithium	Online, cheap, EMF prediction	Needs long rest time (current = 0)
Linear model	Lead photovoltaic	Online, easy	Needs reference data for fitting parameters
Impedance spectroscopy	All systems	Gives information about SoH and quality	Temperature sensitive, cost intensive
DC internal resistance	Lead, Ni/Cd	Gives information about SoH; possibility of online measurements	Good accuracy, but only for a short time interval
Artificial neural networks	All battery systems	Online	Needs training data of a similar battery, expensive to implement
Fuzzy logic	All battery systems	Online	Ask memory in real-word application
Kalman filters	All battery systems, PV, dynamic application	Online; dynamic	Difficult to implement the filtering algorithm that considers all features as, e.g., nonnormalities and nonlinearities

Table 5. Main battery-management IC producers.

Texas Instruments Battery-management ICs	**Philips Semiconductors** Charge-control and monitor ICs	**Microchip Technology Inc.** Battery-management ICs
Hitachi America Ltd Power-management μCs	**Zilog Inc.** μC-based charge controller	**Integrated Circuit Systems Inc.** Charge-control ICs
Linear Technology Corp Battery-management ICs	**Maxim Intergrated Products** Charge-control ICs, SMBus-control ICs	**National Semiconductor Corp**. Battery-management ICs
Analog Devices Battery-management ICs	**Xicor** Battery-management ICs	

Book-keeping Texas Instruments (TI) SoC ICs such as the bq2040, bq26220 [55], bq2060, bq2063 and many others can be found in the market. In 2003, TI announced the release of the bq26500 battery fuel gauge, which for the first time also incorporates an on-board processor to calculate the remaining battery capacity and system run-time (time-to-empty). The device measures the charge and discharge currents using an integrated low-offset voltage-to-frequency converter. A separate ADC on the IC is used to measure battery voltage and temperature. Two count registers accumulate charge and discharge counts, which are generated by sensing the voltage between the two sense resistor pins. Using the measurement inputs, the bq26500 runs a proven book-keeping algorithm (Coulomb counting which is also compensated for self-discharge and discharge rates) to accurately calculate remaining battery capacity and system run-time. The bq26500 compensates remaining battery capacity and run-times for temperature variations. The host system processor simply reads the data set in the bq26500 to retrieve remaining battery capacity, run-time and other critical information that is fundamental to comprehensive battery and power management, including available power, average current, temperature, voltage and time-to-empty and full charge [56]. In the TI ICs, a self-discharge count register counts at a rate of one count every hour at 25 °C [57]. The self-discharge count rate doubles approximately every 10 °C up to 60 °C and is halved every 10 °C below 25 °C down to 0 °C. This value is useful in estimating the battery self-discharge based on capacity and storage temperature conditions.

The Microchip P3 SMBus Smart Battery ICs contain advanced battery control algorithms to determine remaining capacity, run-times and the various other data available to a power-management system [58]. These book-keeping algorithms also rely on a battery cell model that provides the performance information of the particular chemical system in use. The cell models are often referenced as 'look-up tables' or LUTs. The look-up tables are stepwise approximations of the performance response curves that can be drawn when looking at discharge performance as a function of temperature and discharge rate for instance. There are two look-up tables that define the predictive model for the lithium-ion cell chemistry. One look-up table predicts values of residual capacity (the capacity that cannot be removed from the battery due to discharging efficiency) that are used in remaining time calculations. The second is a predictive model of lithium self-discharge. The self-discharge parameter table predicts self-discharge rates as a function of temperature and total capacity loss. Creating look-up tables from cell data requires generating enough test points to create a series of curves which can be broken down into intervals to estimate the curvature with a stepwise approximation. Some LUT 'tables' are illustrated in figure 16.

The top left graphic in figure 16 represents the charge-efficiency model for the NiMH chemistry, the top right represents Li-ion discharge cell models. In the second row

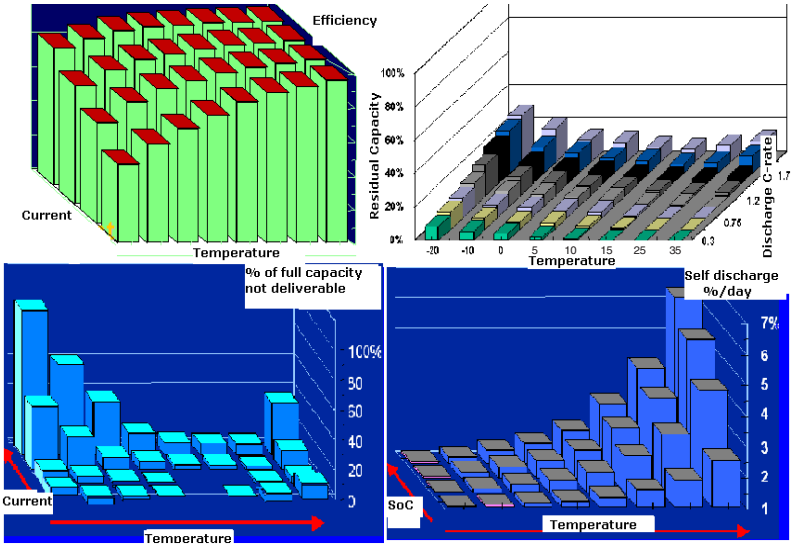


Figure 16. Microchip 3D batteries models [59].

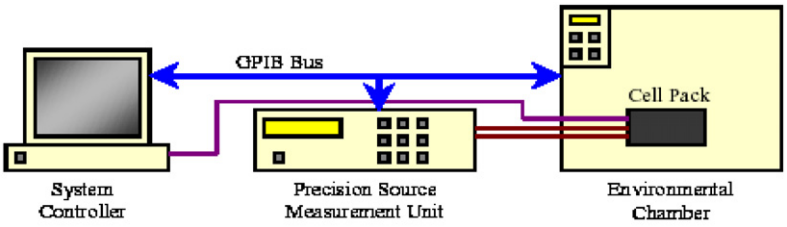


Figure 17. Cell characterization system.

of the figure, the discharge performance and self-discharge in the case of NiMH batteries are represented left and right, respectively.

Maxim battery-management devices for Li-ion cells with a Coulomb counter, temperature converter and 15 bytes of user EEPROM, such as the DS2760 high-precision Li-ion battery monitor or 2438 integrated current accumulator (ICA), can be found in practice. For the Maxim book-keeping algorithms to function accurately while minimizing computational complexity and parametric data storage, certain assumptions are made.

Charge efficiency and pack self-discharge rate are assumed negligible in the case of Li-ion applications and are ignored [45]. The fuel gauging equations work by comparing the ICA value with expected 'empty' and 'full' values for that cell type which are stored in the IC user EEPROM. These data are generated by characterization of the cell type over the expected temperature range and current consumption of the application. This information is subsequently stored in a pack-resident memory for the algorithms to later extract and modify. Figure 17 shows the system used to collect the characterization data. Information should be gathered on several packs so that average or typical values can be stored in every production pack. For best accuracy, the data should be collected on assembled packs containing the production circuit as opposed to individual cells [45].

To collect the data, the cell pack is fully charged at each temperature and fully discharged at each rate at each temperature. All collected data points are arranged as shown in table 6. Since only the difference between points is important, the absolute values of the data do not matter, they have been normalized to the lowest value (standby current empty at 40 °C). This reduces the number of data needed to be stored since standby empty 40 °C is now always 0.

Table 6. Cell characterization data.

	0 °C	10 °C	20 °C	30 °C	40 °C
Full (mA h)	554	561	578	582	588
Standby empty (mA h)	65	42	19	11	0
Active empty (mA h)	124	90	65	50	44

The active empty and standby empty points are defined as the capacity where the battery reaches the empty voltage (as defined by the user) under the load of the active current and standby current, respectively [60]. The characterization data are stored in two pages of the IC EEPROM memory. After characterization of the cell pack is complete, calculating remaining capacity is very simple. The characterization data are used to find the cell full and empty points based on temperature and discharge rate.

The main drawback of all the discussed ICs is that none of them includes an adaptive method, which would allow to deal with spread in both battery and user behaviour, a large temperature and current range and ageing of the cells, under all realistic user conditions.

7. Conclusions

An overview of the state-of-the-art of state-of-charge indication of rechargeable batteries, including a historical development of SoC technologies, has been given in this review. The general operational mechanism of batteries and the characteristics of the best-known three battery types have been given in section 2.

The focus in section 3 has been on the historical development of the SoC indication. A general functional architecture of an SoC system has been described in section 4.

Three basic methods for SoC indication have been identified in section 5, including that of direct measurements. A particular method of direct measurements is the EMF method. The advantage of this presented method is that the EMF curve does not depend on many parameters. It does not depend on ageing and temperature of the battery, so it is a possible candidate for state-of-charge indication. The main drawback of the EMF method is that it does not provide continuous indication of state-of-charge.

A book-keeping system with a particular example for a mobile phone application has been discussed. The main problem in a book-keeping systems is to define reliable calibration opportunities that occur often enough during the battery use [1].

Three of the best-known adaptive methods, which have as inputs battery measured variables such as voltage, impedance and current and use these variables in order to accurately predict the SoC and the remaining time of use for an application have been described. Using at least one of these presented adaptive systems should improve the system's capability to cope with ageing and temperature effects and spread in both battery and user behaviour [1].

The focus in section 6 has been on commercially available SoC systems. In this section a couple of the actual technologies, which the main companies use in practice, have been discussed. The drawback of all the discussed ICs is that none of them includes a real adaptive method, which would allow one to deal with spread in both battery and user behaviour, a large temperature and current range and ageing of the cells, under all realistic user conditions.

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